

# BLUETOOTH SCATTERNETS

## A Cost-Effective Solution for Communication

The most popular application of Bluetooth is communication between two directly paired devices. Here, we explore a less popular but powerful application of Bluetooth which can help extend the range of Bluetooth communication and provide a free infrastructure for communication



■ AHZAM ALI

**B**luetooth is a low-power, low-cost and short-range wireless technology. It was originally developed by Ericsson for short-range communication between personal devices, e.g., data transfer between a cellphone and a computer, or communication with a printer. Bluetooth devices use the unlicensed ISM band at 2.4 GHz.

Nowadays most of the mobile devices, including mobile phones, MP3 players, digital cameras, video cameras, PDAs, laptop PCs, tablet PCs and so forth, are equipped with Bluetooth. So you can exchange data between these devices without requiring the use of wires. Bluetooth is considered to be a high-potential technology for providing wireless communication in a home-networking environment.

Bluetooth is an evolving technology. It has gone through three (standardised) revisions of the approved standard by Bluetooth Special Interest Group (SIG), the latest one being Bluetooth 2.1 EDR (enhanced data rate) in early part of 2008. The latest standard supports enhanced usability, i.e., pairing of devices by means of near-field communication (NFC), enhanced security and broader range of device profiles.

When a number of Bluetooth devices communicate to each other in the same vicinity, there is a high level of interference. To combat interference, Bluetooth technology applies a fast frequency-hopping scheme which hops over 79 channels 1600 times per second. For devices to communicate to each other using Bluetooth they need to be paired with each other to have synchronised frequency-hopping sequence.

### Scatternets: A set of piconets

When there is a collection of devices paired with each other, it forms a small personal area network called 'piconet.' A piconet consists of a master and at most seven active slaves. Each piconet has its own hopping sequence and the master and all slaves share the same channel. In a piconet, the master and slave devices transmit packets in even and odd slots, respectively.

Two or more piconets connected to each other by means of a device (called 'bridge') participating in both the piconets, form a scatternet. The role of the bridge is to transmit data across piconets. Fig. 1 shows a scatternet structure in which devices 'a' and 'c' play the master role and device 'b' acts as the bridge.

The scatternet formation has not been formally defined in the Bluetooth SIG specifications. As a result, numerous protocols have been proposed. While modeling *ad-hoc* networking, in general, is complex, the additional restrictions imposed by the Bluetooth specifications—such as low cost of the device, low power consumption and network resilience while using piconets that have a maximum of seven active nodes—have created a

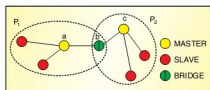


Fig. 1: Scatternet structure